

CIPD 360 ERP – Complete Feature Breakdown

1. User Management & Authentication System

This is the foundation of the system.

1.1 Role Based Login

Three primary user roles:

Admin

- Full system access
- Scheduling control
- Attendance correction
- Reporting and analytics
- Faculty payment management

Faculty

- View teaching schedule
- Upload lecture content
- View student attendance
- View aggregated feedback
- Grade assignments

Student

- View class schedule
- Attend sessions
- Submit feedback
- Access learning resources
- Upload assignments

This structure is explicitly mentioned in both the SRS and meeting discussion.

1.2 Authentication Features

- Email + password login
- Secure password hashing
- Password reset via email
- Session tokens (JWT)
- Session timeout
- Logout functionality

1.3 Role-Based Access Control (RBAC)

Different permissions for:

Feature	Admin	Faculty	Student
Scheduling	✓	View	View
Attendance override	✓	View	View
Upload content	✓	✓	View
Feedback analytics	✓	✓	Submit
Reports	✓	Limited	Limited

2. Dashboard System

Each role gets a **separate dashboard**.

2.1 Admin Dashboard

Admin needs a **complete operational overview**.

Features:

- Attendance statistics
- Student participation
- Faculty workload
- Feedback summary
- Upcoming sessions
- Recent uploads
- Honorarium calculation summary

Visual analytics:

- Attendance graphs
- Feedback distribution
- Session counts
- Weekly participation

Charts suggested in meeting discussion.

2.2 Faculty Dashboard

Faculty should see:

- Today's sessions
 - Weekly teaching schedule
 - Uploaded materials
 - Student attendance stats
 - Feedback analytics for their sessions
 - Assignment submissions
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2.3 Student Dashboard

Students see:

- Upcoming sessions
 - Class schedule
 - Attendance percentage
 - Feedback pending alerts
 - Uploaded lecture material
 - Assignment submission panel
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3. Scheduling Management System

Scheduling is one of the **core systems**.

Admin creates sessions.

3.1 Session Creation

Admin can specify:

- Session title / topic
- Faculty name
- Date
- Time
- Duration
- Venue
- Course

Mentioned directly in meeting discussion.

3.2 Multiple Sessions per Day

System must support:

- 4 sessions typical
- Expandable to 5–6 sessions

Admin can **add sessions dynamically using "+" button**.

3.3 Recurring Sessions

Admin can create:

- Daily schedules
 - Weekly schedules
 - Manual schedule entries
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3.4 Conflict Detection

System detects:

- Faculty schedule clash
- Venue conflict
- Student schedule conflict

Admin can override conflicts with justification.

3.5 Automatic Schedule Distribution

When session is created:

- Appears on student dashboard
- Appears on faculty dashboard
- Email reminders sent

Meeting transcript explicitly mentions **automatic dashboard updates and email reminders**.

3.6 Calendar Integration

Features:

- Calendar view
 - List view
 - Email calendar invites (.ics)
-

4. Attendance Management System

This is the **most technically unique feature of the system**.

Attendance is automated using **Wi-Fi router detection**.

4.1 WiFi Based Attendance

Logic:

1. Student connects to campus WiFi
2. Router BSSID is captured
3. System logs device hash
4. Ping occurs periodically

Attendance rule:

- **5 pings per hour**
- **12 minute interval**
- **≥3 detections = PRESENT**

Defined clearly in SRS.

4.2 Attendance Window

Attendance is tied to session timing.

Example:

Session: 3pm–4pm

Pings:

3:00

3:12

3:24

3:36

3:48

If ≥3 detected → present.

4.3 Attendance Data Storage

System stores:

- device hash
- BSSID
- timestamp

- session id

Stored for auditing.

4.4 Attendance Analytics

Admin and students can view:

- Daily attendance
- Weekly trends
- Monthly statistics

Graphs suggested in meeting discussion.

4.5 Manual Attendance Correction

Admin can:

- modify attendance
 - correct false absence
 - maintain correction history
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5. Feedback System

Feedback is collected **after every session**.

5.1 Session Based Feedback

Each class triggers feedback request.

Students rate:

- Structure
- Content
- Logistics & Support

Mentioned in meeting transcript.

5.2 Feedback Question Types

Supported formats:

- Rating scale (1–5)
- Yes / No questions

- Conditional questions

Example:

Did class start on time?

Yes / No

If No → What issue occurred?

Defined in SRS.

5.3 Feedback Completion Constraint

Feedback must be **completed within ~60 seconds**.

Design goal:

- quick
- high completion rate

Mentioned in SRS.

7. Content Management System

This acts like a **lightweight LMS (similar to Google Classroom)**.

Mentioned in discussion.

7.1 Lecture Material Upload

Faculty can upload:

- PPT
 - PDF
 - Links
 - Notes
 - External resources
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7.2 Session Based Content

Content is linked to specific session.

Example:

Session:

Machine Learning Basics

Content:

Lecture Slides

Reference PDF

7.3 Upload Permissions

Uploads allowed by:

- Faculty
- Admin

Students only download.

8. Assignment & Submission System (Discussed but Lower Priority)

Future extension.

8.1 Assignment Upload

Faculty can post:

- Assignments
 - Projects
 - Homework
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8.2 Student Submission

Students upload:

- PDFs
 - Documents
 - Reports
-

8.3 Submission Metadata

Stored:

- student ID
 - submission timestamp
 - file version
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8.4 Grading System

Faculty can:

- evaluate assignments
- give marks
- attach comments

Mentioned in transcript discussion.

9. Faculty Workload & Honorarium System

Important administrative feature.

9.1 Faculty Session Tracking

System records:

- faculty name
- sessions conducted
- total hours

Mentioned explicitly in meeting.

9.2 Monthly Honorarium Calculation

Admin can generate report:

Faculty | Sessions | Hours | Payment

Exportable.

10. Notification System

Automated communication.

10.1 Email Notifications

Triggers:

- class reminder
- schedule change
- missing feedback
- attendance warning

Defined in SRS.

10.2 Reminder System

Example:

Student missed feedback.

Email:

"Please submit feedback for today's session."

Mentioned in meeting discussion.

11. Reporting System

Analytics for decision making.

11.1 Admin Reports

- Attendance statistics
 - Feedback trends
 - Faculty workload
 - Session analytics
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11.2 Export Reports

Formats:

- CSV
- PDF

Defined in SRS.

12. Student Profile System

Discussed in meeting.

Each student profile contains:

- personal info
- enrolled program
- attendance history
- feedback history
- assignment submissions

Mentioned in transcript.